

KPAI: Engineering Trustworthy and Impactful AI Systems

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Abstract—The technological landscape of 2026 is characterized by the GenAI Divide, a phenomenon where approximately 95% of generative AI pilots fail to transition into productive assets that deliver measurable financial impact. This failure is primarily attributed to a measurement gap where operational telemetry fails to explain shifts in business value or enterprise profit and loss. This solution concept paper contributes the KPAI framework, providing a systematic approach to requirements engineering for trustworthy and impactful AI systems. KPAI enables Chief AI Officers to transform unpredictable AI implementations into disciplined, strategic assets that foster organizational trust. The managed discipline for AI evolution is activated through three distinct phases. Clarity phase facilitates automated asset discovery and the creation of an impact topology map to visualize dependencies across complex multiagent ecosystems. Consistency phase establishes quality requirements baselines by utilizing golden benchmarks and a metric matcher to align technical performance with deterministic business outcomes. Confidence phase employs evolution sensitivity to ensure changes remain economically viable and compliant with regulatory mandates. The framework is demonstrated on a real-world case study.

Keywords— *Trustworthy AI, AI Quality KPI, AI Impact, AI Change and Control, AI Risks, AI Validation.*

I. INTRODUCTION

The technological landscape of 2026 is defined by a paradox where immense capital investment in AI exists alongside a pervasive failure to realize tangible financial returns. While collective corporate expenditure on generative AI has exceeded \$40 billion annually, approximately 95% of generative AI pilots fail to transition into productive assets that deliver measurable core operating case profit [5]. This phenomenon, identified by the Massachusetts Institute of Technology as the GenAI Divide [1], underscores a critical deficiency in the application of engineering discipline to AI systems. Boards demand rapid acceleration, CFOs require documented financial impact, and regulators enforce strict deterministic control.

A. Business Adoption Status

Omdia [7] surveyed 2,050 professionals worldwide between Aug. 13, 2025 and Sept. 17, 2025. That survey indicated that 32% of respondents say they have agentic solutions in production and 40% respondents quantified their Return on Investment (ROI) on gen AI report earning \$1.49 for every \$1 invested. Yet, in this survey executives anticipated 41% project

failure rate which highlights a substantial gap between leadership optimism and technical reality. Quality reported challenges were 40% regarding data, 31% regarding integration with existing or legacy systems, and 27% regarding scalability and performance. The key use cases were 51% for operational efficiency that should be measured with business Key Performance Indicators (KPI). These statistics might indicate that it is not a matter of experiencing an AI failure, rather experiencing a measurement failure [2] with lack of clear business KPI mapped to responsible AI agent. Existing trustworthy AI frameworks [3] articulate transparency, accountability, and technical robustness as foundational properties; however, they do not prescribe how these properties are continuously verified in live, evolving agentic systems, the operational gap this work addresses.

B. Problem Definition

The challenge of establishing business trust in AI systems is fundamentally a problem of quality requirements, including integration, data ingress, scale, performance and efficiency. Trustworthy AI is not merely a technical attribute but a human-centered construct that must be maintained over the entire system lifetime from design to production. This encompasses technical performance, business and societal implications, and rigorous adherence to ethical and legal frameworks such as the European Union AI Act [3].

For business leaders and Chief AI Officers (CAIO), trust in AI systems is predicated on three essential business needs throughout the AI systems lifecycle:

- Clarity in integrated AI components fit to business goals. Key questions are:
 - What AI assets do I have?
 - What contribution do they generate once deployed?
- Consistency in performance and data processing with expected efficiency. Key questions are:
 - How should I evaluate AI systems trust?
 - How can I measure business impact and performance?
- Confidence in systematic and predicted AI evolution for a long-term success. Key questions are:

- When and how do I decide on an AI system evolution?
- When do we achieve economic viability?

This practitioner paper presents a concept and preliminary state of the KPAI framework. KPAI platform for continuous AI validation increases trust in AI and measure its impact on the Profit and Loss (P&L). The paper presents a set of tooling for setting AI quality requirements to address the clarity, consistency and confidence needs of the CAIOs. From automatic discovery of AI assets, measurements of business KPIs, to the evaluation of AI components alternatives for increasing efficiency and other quality attributes of new and existing AI systems, KPAI addresses the 3 key business trustworthiness needs of CAIOs.

II. THE KPAI FRAMEWORK FOR CONTINUOUS AI VALIDATION

A. Solution Rationale

AI-driven business value and ROI is expressed by business KPIs. Yet there is a distinct measurement gap between operational telemetry and the realization of enterprise value, particularly in the context of AI investments. While useful for monitoring system health, operational metrics, fail to explain shifts P&L statements, business value and impact. Existing observability frameworks and standard such as OpenTelemetry¹ (OTel) metrics are implemented by Application Performance Monitoring (APM) tools. An example is Amazon Bedrock AgentCore², that via CloudWatch can monitor and evaluate agents in production and gain operational insights and quality assurance with real-time visibility into AI assets performance. CloudWatch can measure metrics such as token usage, session duration, and error rates to evaluate technical AI assets quality including correctness and goal success rate. Yet such APM tools provide little insight into whether operational efficiencies have translated into business gains requirements such as customer lifetime value, churn rate, or net profit margins. These tools and standards are designed for high-fidelity technical tracing and monitoring but lack the semantic structures to represent business logic or financial impact.

This gap necessitates a transition to an industrialized mandate where requirements engineering (RE) facilitates the alignment of technical variations with deterministic business outcomes. New frameworks exist that evaluate AI agents across both technical and financial domains [4], proposing dependencies between technical and financial metrics, yet lack a comprehensive quality KPIs and needed improvement for the entire lifecycle of agentic and AI systems for continues improvement. The industrialization of AI tools is further complicated by the rise of Multiagent Systems (MAS) and the Agentic Web [8]. By 2026, technology trends have shifted away from generic Large Language Models toward Domain-Specific Language Models (DSLMS) and complex multiagent ecosystems that collaborate on shared goals [5]. These systems require a shift in RE from managing individual models to orchestrating agentic workflows and complex AI assets dependencies from AI business services to technical components. Hence, due to scale, variety and complexity of

managed AI assets and the velocity and magnitude of changes these AI assets undergo, ensuring trust in business outcome across the AI portfolio, is an elusive target. Any modification such as upgrading a model, adjusting a tool, or integrating a new agent, triggers full compliance obligations (the definition of a substantial modification is provided in Art. 3(23) of the EU AI Act). This creates a set of AI portfolio requirements for CAIOs that need to maintain clarity over a living portfolio of AI assets and their topological dependency and relations, as an audit trail that tags assets versions to every business output. This necessitates a disciplined and continuously validated RE practice, particularly with respect to AI asset lineage and linking AI outputs to measurable business outcomes for stakeholders, customers, and regulators.

B. The KPAI framework phases and technology

The KPAI framework addresses the fragmented landscape of AI management by defining AI evolution and control as a managed engineering discipline based on ensuring clarity, consistency, and confidence requirements are set and met. This framework automates the discovery, topology, analysis, and improvement of the AI value chain through three distinct phases. It provides contextual validation, augmenting technical scores with KPI-centric scoring that measures the impact of AI changes on real business metrics such as lead conversion rates or procurement cycle times.

1) *Phase 1 - **Clarity** and the Mapping of the AI Topology:* Achieving clarity requires an understanding of the organization's existing AI assets. The *AI Asset Discovery* capability detects AI assets such as foundation models, agents, tools, providers, skills and more with static code analysis or detected in run time via OTel instrumentations and APM once third-party tools are activated. These AI assets are linked by the *Process and AI Inventory* module, to detect where AI assets are part of the business processes. Overall, the result is an *Impact Topology Map*, that is an internal catalog, in the form of a knowledge graph that contains AI assets in the context of business processes and applications. For example, an analysis of this topology can deduce that a change in a process ingress component, such as entity extraction, can cause failures in a process egress component, such as financial reporting. Mapping this impact radius across managed processes and AI assets ensures that stakeholders can visualize the consequences before they reach production and ensure trace log for audit.

2) *Phase 2 - Establishing **Consistent Quality**:* The second phase focuses on maintaining business trust and performance of a managed AI systems over time. This involves the use of *Golden Benchmarks*, which are ground truth datasets curated from historical transactional data. The benchmark enables retrospective analysis when model behavior drifts or a new variant is introduced to ensure evaluations remain deterministic and reproducible across the full asset lifecycle. These benchmarks serve as the evaluation testing set for all new AI variations, to ensure performance remains stable despite updates to models or agents. A key component is the *Metric*

¹ <https://opentelemetry.io/>

² <https://aws.amazon.com/bedrock/>

Matcher, which maps technical metrics such as accuracy and latency to consistent business quality requirements and KPIs. Table I and table II contain examples for such metrics and KPIs in the context of Ingress (pre) and Egress (post) data processing with AI assets. This allows organizations to establish a quality harness, that is contextual baseline for verifying AI system trust.

3) *Phase 3 - **Confident** Evolution Change and Playbooks*: The final phase transforms the framework into a decision engine, providing evidence for evolution actions such as scaling, fixing, improving, or retiring AI assets. The *Variation Engine* generates configuration and changes recommendations, such as swapping assets. It also provides technical change requirements in the form of instructions and specifications, code snippets or prompts in natural language for developers. This phase relies on the *Alternative Scout* module, which identifies external (global third party) or internal (active) AI assets that could replace ones that are under evolution change. The selected alternative for evolution should support and enhance the required business quality attributes such as performance-to-cost ratios and ROI. Phase 3 also includes a comparison between variation and existing AI systems for selecting the best alternative.

TABLE I. INGRESS AI COMPONENTS

<i>AI Component (Ingress)</i>	<i>Functional Description</i>	<i>OTel Metric</i>	<i>Inferred Business Requirements KPI</i>
Entity Extraction	Pulling names, dates, amounts.	F1-Score / % of error vs success of activation.	Manual Claim Processing Cost (Errors trigger manual review).
Document Classification	Sorting invoices from legal files.	Categorization Precision.	Process Automation Rate (Misclassification stalls RPA).
Intent Triage	Sorting tickets by Refund vs. Query.	Routing Accuracy.	Lead Conversion (Delays decrease sales velocity).

TABLE II. EGRESS AI COMPONENTS

<i>AI Component (Egress)</i>	<i>Functional Description</i>	<i>OTel Metric</i>	<i>Business KPI</i>
Policy Compliance	Checking output against rules.	Compliance Violation Count.	Legal Review Latency (Forces human sign-off).
Risk Scoring	Assigning risk to a response.	Severity Distribution.	Operational Risk Multiplier (Cost of error).
Tone Adjustment	Formal to Empathetic shifts.	Sentiment Consistency.	CSAT (Loyalty drop due to robotic tone).

C. Example - Entity Extraction AI Asset Metrics Mappings

A rigorous business RE discipline for AI requires bridging the gap between technical telemetry and business impact. The examples in tables I and II map core ingress and egress components to standard OTel metrics and their corresponding business KPI impacts. One example is the *Entity Extraction* component, that can pull names, dates and amounts out of an unstructured document. The technical metric can be a ratio of accuracy versus ground truth or error and fail ratio related to total activations, yet the business requirement is derived from a specific business service goal. For example, consider a claim processing AI agent, with the goal to extract information from a mail and process an insurance claim. If there is a failure in *Entity Extraction* ingress component, human labor needs to replace the automated one. Hence, the business requirement should limit the amount of labor hours according to an economical equilibrium as detailed in equation (1) to (4). This labor is measured by a KPI of Manual Claim Processing Cost, that is correlated with the error rate and multiplied by the transaction volume of claims in a period and by the average cost to handle a single claim manually.

III. DEMONSTRATION OF APPLICABILITY

The EU AI Act Art. 43(4) and Art. 25(1)(b–c) dictates that any significant modification to an AI system triggers full compliance obligations. This creates a requirement for Chief AI Officers (CAIOs) to maintain clarity over a live topology and an audit trail log that tags model versions to every output, such as depicted in figure 1. This transforms legal burdens into a disciplined engineering requirement, ensuring that the organization can prove its compliance posture to regulators at any time. However, the decision to evolve an AI system by swapping AI assets must be grounded in economic reality. A technical optimization, such as moving to a cheaper model, that needs to maintain compliance, also may inadvertently increase human labor costs due to a drop in accuracy. As an example, the KPAI framework applies a sensitivity score to calculate viability based on technical observability metrics and labor costs, according to set requirements quality levels.

A. Calculation of Evolution Sensitivity

Define the following variables:

C_{AI_base} = Unit cost of the existing AI system or asset.

C_{AI_new} = Unit cost of the new AI system or asset.

V = Transaction volume.

L_h = Cost of human labor per manual alternative operation.

A_{base} = Accuracy rate of the existing AI system or asset.

A_{new} = Accuracy rate of the new AI system or asset.

The Total Cost of Ownership (TCO) KPI for the base and new configurations are defined as:

$$TCO_{base} = V \times \{ C_{AI_base} + L_h \times (1 - A_{base}) \} \quad (1)$$

$$TCO_{new} = V \times \{ C_{AI_new} + L_h \times (1 - A_{new}) \} \quad (2)$$

An evolution is recommended only when the economics are tangible [6]:

$$TCO_{base} - TCO_{new} > 0 \quad (3)$$

Substitution of the TCO equations 1 and 2 into 3, indicates that a change is recommended only when the average cost of

human labor (L_h) is less than the ratio of the change in cost over the change in accuracy:

$$L_h < (C_{A_base} - C_{AI_new}) / (A_{base} - A_{new}) = \Delta C / \Delta A \quad (4)$$

This Evolution Sensitivity demonstrates a coupling of none-functional requirements, defining an inferred business KPI for evolution represented in equation 4.

B. Evolution Sensitivity of an Insurance Claims Case Study

In an investigation of an automated insurance claims process, an organization considered swapping GPT-4o for Llama-3-70B to reduce token costs. The existing system had a cost of \$0.05 per completion with 98.2% accuracy. The new system cost was \$0.008 with a reduced accuracy of 94.5%, a 3.7% drop.

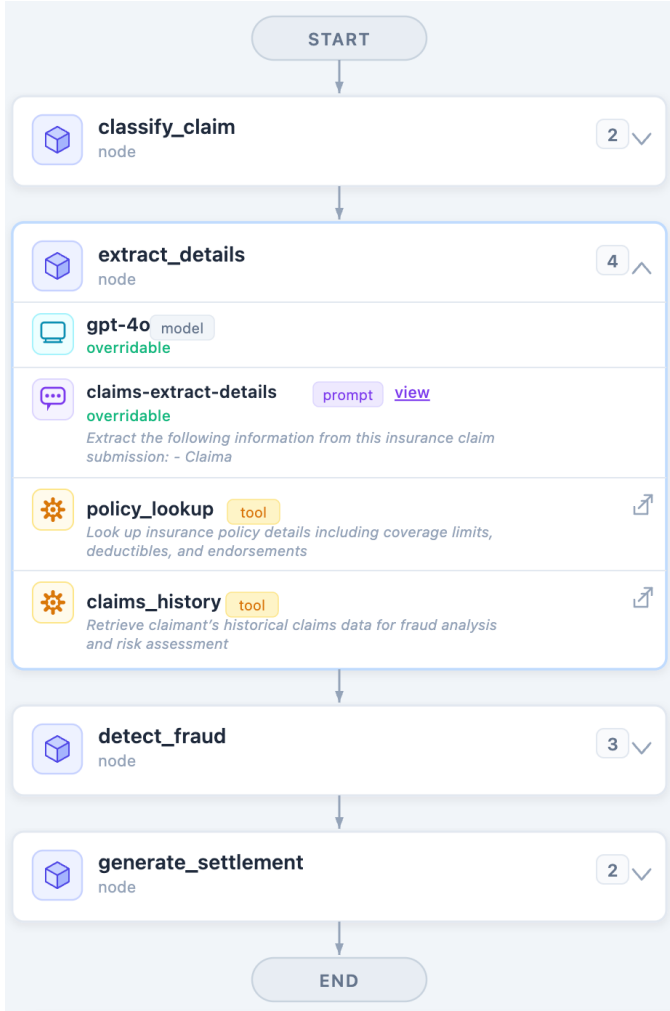


Fig. 1. Insurance Claim Agentic Processing Audit Trail Log

While the new model reduced direct AI costs by over 80%, and was still a valid EU AI act compliant model, the surge in manual review hours (estimated at 20 minutes of labor per error) made the cheaper model significantly more expensive for the enterprise. The validation confirms that if technical accuracy

drops even marginally, the labor cost can outweigh any technical savings. The Confidence Threshold, which is the accuracy drop that still ensure that the evolution remains profitable, was found to be 1.8% in this specific case. This highlights the need for additional inferred business requirement for Evolution Sensitivity, with set KPI threshold, controlling the AI changes alternatives selection.

Prior work on business-driven IT management and AI observability has explored aligning technical metrics with economic outcomes [6]. KPAI extends these efforts by combining requirements engineering discipline with a KPI-centric scoring harness that spans the full AI asset lifecycle, from automated discovery to sanctioned evolution, a scope not addressed by existing platforms.

IV. THE LANDSCAPE OF AI MANAGEMENT PLATFORMS

Modern organizations utilize various platforms to manage their AI assets, yet many present significant gaps in evolution control. The current market is fragmented across lifecycle approval, regulatory mapping, and technical observability. For example, ServiceNow AI Control Tower³ strength is its lifecycle approval and asset visibility, however it lacks technical emulation and node-level mapping. OneTrust⁴ AI governance regulatory mapping and risk tiering, predominately focus on legality over performance or profitability. Lastly, Arize AI⁵ observability technical tracing and AI drift detection, lacks business stakeholder simulation and ROI mapping. While ServiceNow excels at the process of approving solutions and OneTrust leads in risk tiering, they lack the technical depth to emulate how specific AI component changes (as shown in Fig. 1), will impact adhering to a business requirement, measured via KPIs at a technical node level. Conversely, Arize provides technical leadership in drift detection but lacks the simulation workflows needed for business stakeholders to evaluate ROI at a portfolio level to match business requirements, goals and constraints.

V. CONCLUSION

The 2026 AI ROI reckoning marks the transition of AI development from an art form to a managed engineering discipline. CAIOs who fail to implement a phased roadmap of clarity and consistency, will find their budgets restricted by boards no longer tolerant of invisible earning power. By adopting frameworks that map technical variations to contextual requirements manifested as business KPIs, enterprises can finally bridge the GenAI Divide with trusted AI for businesses. This work-in-progress of the KPAI framework and solution concept aims to support these needs by defining and measuring AI business quality requirements for increasing business trust with AI systems, and consequently proposed changes, alongside alternative design to cultivate and ensure business trust with AI.

FUTURE WORK

KPAI is a commercial startup product and will not be released as open source. Its strategic direction is to serve as the full control plane for enterprise agentic workflow management.

³ <https://www.servicenow.com/products/ai-control-tower.html>

⁴ <https://www.onetrust.com/>

⁵ <https://arize.com/>

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